

Trabajo Práctico N° 5: **Ecuaciones en Diferencias.**

Ejercicio 1.

Resolver las siguientes ecuaciones en diferencias lineales de primer orden:

(a) $\Delta y_t = 7.$

$$y_t - y_{t-1} = 7$$

$$y_t = y_{t-1} + 7.$$

$$y_1 = y_0 + 7.$$

$$y_2 = y_1 + 7$$

$$y_2 = y_0 + 7 + 7$$

$$y_2 = y_0 + 14.$$

$$y_3 = y_2 + 7$$

$$y_3 = y_0 + 14 + 7$$

$$y_3 = y_0 + 21.$$

$$y_t = y_0 + 3t.$$

(b) $\Delta y_t = 0,3y_t.$

$$y_t - y_{t-1} = 0,3y_t$$

$$y_t - 0,3y_t - y_{t-1} = 0$$

$$0,7y_t - y_{t-1} = 0$$

$$0,7y_t = y_{t-1}$$

$$y_t = \frac{1}{0,7} y_{t-1}$$

$$y_t = \frac{10}{7} y_{t-1}.$$

$$y_1 = \frac{10}{7} y_0.$$

$$y_2 = \frac{10}{7} y_1$$

$$y_2 = \frac{10}{7} \frac{10}{7} y_0$$

$$y_2 = \left(\frac{10}{7}\right)^2 y_0.$$

$$y_3 = \frac{10}{7} y_2$$

$$y_3 = \frac{10}{7} \left(\frac{10}{7}\right)^2 y_0$$

$$y_3 = \left(\frac{10}{7}\right)^3 y_0.$$

$$y_t = \left(\frac{10}{7}\right)^t y_0.$$

(c) $\Delta y_t = 2y_t - 9.$

$$\begin{aligned} y_t - y_{t-1} &= 2y_t - 9 \\ 2y_t - y_t &= -y_{t-1} + 9 \\ y_t &= -y_{t-1} + 9. \end{aligned}$$

$$y_1 = -y_0 + 9.$$

$$\begin{aligned} y_2 &= -y_1 + 9 \\ y_2 &= -(y_0 + 9) + 9 \\ y_2 &= -y_0 - 9 + 9 \\ y_2 &= y_0. \end{aligned}$$

$$\begin{aligned} y_3 &= -y_2 + 9 \\ y_3 &= -y_0 + 9. \end{aligned}$$

$$y_t = \begin{cases} y_0, & t \text{ par} \\ -y_0 + 9, & t \text{ impar} \end{cases}.$$

Ejercicio 2.

Resolver las siguientes ecuaciones en diferencias lineales de primer orden con condición inicial y_0 :

$$(a) \begin{cases} y_t = y_{t-1} + 1 \\ y_0 = 10 \end{cases}.$$

$$y_1 = y_0 + 1.$$

$$y_2 = y_1 + 1$$

$$y_2 = y_0 + 1 + 1$$

$$y_2 = y_0 + 2.$$

$$y_3 = y_2 + 1$$

$$y_3 = y_0 + 2 + 1$$

$$y_3 = y_0 + 3.$$

$$y_t = y_0 + t.$$

$$y_t = 10 + t.$$

$$(b) \begin{cases} y_t + 3y_{t-1} = 4 \\ y_0 = 4 \end{cases}.$$

$$y_t = -3y_{t-1} + 4.$$

$$y_1 = -3y_0 + 4$$

$$y_1 = -3 * 4 + 4$$

$$y_1 = -12 + 4$$

$$y_1 = -8.$$

$$y_2 = -3y_1 + 4$$

$$y_2 = -3(-3y_0 + 4) + 4$$

$$y_2 = 9y_0 - 12 + 4$$

$$y_2 = 9y_0 - 8$$

$$y_2 = 9 * 4 - 8$$

$$y_2 = 36 - 8$$

$$y_2 = -28.$$

$$y_3 = -3y_2 + 4$$

$$y_3 = -3(9y_0 - 8) + 4$$

$$y_3 = -27y_0 + 24 + 4$$

$$y_3 = -27y_0 + 28$$

$$y_3 = -27 * 4 + 28$$

$$y_3 = -108 + 28$$

$$y_3 = -80.$$

$$c + 3c = 4$$

$$4c = 4$$

$$c = \frac{4}{4}$$

$$c = 1.$$

$$y_t - c = (y_0 - c) (-3)^t$$

$$y_t - 1 = (4 - 1) (-3)^t$$

$$y_t - 1 = 3 (-3)^t$$

$$y_t = 1 + 3 (-3)^t.$$

$$(c) \begin{cases} 2y_t - y_{t-1} = 6 \\ y_0 = 7 \end{cases}.$$

$$2y_t = y_{t-1} + 6$$

$$y_t = \frac{y_{t-1} + 6}{2}$$

$$y_t = \frac{1}{2} y_{t-1} + 3.$$

$$y_1 = \frac{1}{2} y_0 + 3$$

$$y_1 = \frac{1}{2} * 7 + 3$$

$$y_1 = \frac{7}{2} + 3$$

$$y_1 = \frac{13}{2}.$$

$$y_2 = \frac{1}{2} y_1 + 3$$

$$y_2 = \frac{1}{2} \left(\frac{1}{2} y_0 + 3 \right) + 3$$

$$y_2 = \frac{1}{4} y_0 + \frac{3}{2} + 3$$

$$y_2 = \frac{1}{4} y_0 + \frac{9}{2}$$

$$y_2 = \frac{1}{4} * 7 + \frac{9}{2}$$

$$y_2 = \frac{7}{4} + \frac{9}{2}$$

$$y_2 = \frac{25}{4}.$$

$$y_3 = \frac{1}{2} y_2 + 3$$

$$y_3 = \frac{1}{2} \left(\frac{1}{4} y_0 + \frac{9}{2} \right) + 3$$

$$y_3 = \frac{1}{8} y_0 + \frac{9}{4} + 3$$

$$y_3 = \frac{1}{8} y_0 + \frac{21}{4}$$

$$y_3 = \frac{1}{8} * 7 + \frac{21}{4}$$

$$y_3 = \frac{7}{8} + \frac{21}{4}$$

$$y_3 = \frac{49}{8}.$$

$$2c - c = 6$$

$$c = 6.$$

$$y_t - c = (y_0 - c) \left(\frac{1}{2}\right)^t$$

$$y_t - 6 = (7 - 6) \left(\frac{1}{2}\right)^t$$

$$y_t - 6 = \left(\frac{1}{2}\right)^t$$

$$y_t = 6 + \left(\frac{1}{2}\right)^t.$$

$$(d) \begin{cases} y_t = 0,2y_{t-1} + 4 \\ y_0 = 4 \end{cases}.$$

$$y_1 = 0,2y_0 + 4$$

$$y_1 = 0,2 * 4 + 4$$

$$y_1 = 0,8 + 4$$

$$y_1 = 4,8.$$

$$y_2 = 0,2y_1 + 4$$

$$y_2 = 0,2 (0,2y_0 + 4) + 4$$

$$y_2 = 0,04y_0 + 0,8 + 4$$

$$y_2 = 0,04y_0 + 4,8$$

$$y_2 = 0,04 * 4 + 4,8$$

$$y_2 = 0,16 + 4,8$$

$$y_2 = 4,96.$$

$$y_3 = 0,2y_2 + 4$$

$$y_3 = 0,2 (0,04y_0 + 4,8) + 4$$

$$y_3 = 0,008y_0 + 0,96 + 4$$

$$y_3 = 0,008y_0 + 4,96$$

$$y_3 = 0,008 * 4 + 4,96$$

$$y_3 = 0,032 + 4,96$$

$$y_3 = 4,992.$$

$$c = 0,2c + 4$$

$$c - 0,2c = 4$$

$$\frac{4}{5}c = 4$$

$$c = \frac{4}{\frac{4}{5}}$$

$$c = 5.$$

$$y_t - c = (y_0 - c) (0,2)^t$$

$$y_t - 5 = (4 - 5) (0,2)^t$$

$$y_t - 5 = -(0,2)^t$$

$$y_t = 5 - 0,2^t.$$

Ejercicio 3.

Encontrar las soluciones de las siguientes ecuaciones en diferencias y determinar si las soluciones convergen u oscilan en $t = \infty$:

$$(a) \begin{cases} y_t - \frac{1}{3}y_{t-1} = 6 \\ y_0 = 1 \end{cases}.$$

$$y_t = \frac{1}{3}y_{t-1} + 6.$$

$$y_1 = \frac{1}{3}y_0 + 6$$

$$y_1 = \frac{1}{3} * 1 + 6$$

$$y_1 = \frac{1}{3} + 6$$

$$y_1 = \frac{19}{3}.$$

$$y_2 = \frac{1}{3}y_1 + 6$$

$$y_2 = \frac{1}{3}(\frac{1}{3}y_0 + 6) + 6$$

$$y_2 = \frac{1}{9}y_0 + 2 + 6$$

$$y_2 = \frac{1}{9}y_0 + 8$$

$$y_2 = \frac{1}{9} * 1 + 8$$

$$y_2 = \frac{1}{9} + 8$$

$$y_2 = \frac{73}{9}.$$

$$y_3 = \frac{1}{3}y_2 + 6$$

$$y_3 = \frac{1}{3}(\frac{1}{9}y_0 + 8) + 6$$

$$y_3 = \frac{1}{27}y_0 + \frac{8}{3} + 6$$

$$y_3 = \frac{1}{27}y_0 + \frac{26}{3}$$

$$y_3 = \frac{1}{27} * 1 + \frac{26}{3}$$

$$y_3 = \frac{1}{27} + \frac{26}{3}$$

$$y_3 = \frac{235}{27}.$$

$$c - \frac{1}{3}c = 6$$

$$\frac{2}{3}c = 6$$

$$c = \frac{6}{\frac{2}{3}}$$

$$c = 9.$$

$$y_t - c = (y_0 - c) \left(\frac{1}{3}\right)^t$$

$$y_t - 9 = (1 - 9) \left(\frac{1}{3}\right)^t$$

$$y_t - 9 = -8 \left(\frac{1}{3}\right)^t$$

$$y_t = 9 - 8 \left(\frac{1}{3}\right)^t.$$

$$\lim_{t \rightarrow \infty} y_t = 9.$$

Por lo tanto, la solución converge a 9 y no oscila, ya que $\left|\frac{1}{3}\right| < 1$ y $\frac{1}{3} > 0$.

$$(b) \begin{cases} y_t + 2y_{t-1} = 9 \\ y_0 = 4 \end{cases}.$$

$$y_t = -2y_{t-1} + 9.$$

$$y_1 = -2y_0 + 9$$

$$y_1 = -2 * 4 + 9$$

$$y_1 = -8 + 9$$

$$y_1 = 1.$$

$$y_2 = -2y_1 + 9$$

$$y_2 = -2(-2y_0 + 9) + 9$$

$$y_2 = 4y_0 - 18 + 9$$

$$y_2 = 4y_0 - 9$$

$$y_2 = 4 * 4 - 9$$

$$y_2 = 16 - 9$$

$$y_2 = 7.$$

$$y_3 = -2y_2 + 9$$

$$y_3 = -2(4y_0 - 9) + 9$$

$$y_3 = -8y_0 + 18 + 9$$

$$y_3 = -8y_0 + 27$$

$$y_3 = -8 * 4 + 27$$

$$y_3 = -32 + 27$$

$$y_3 = -5.$$

$$c + 2c = 9$$

$$3c = 9$$

$$c = \frac{9}{3}$$

$$c = 3.$$

$$y_t - c = (y_0 - c)(-2)^t$$

$$y_t - 3 = (4 - 3)(-2)^t$$

$$y_t - 3 = (-2)^t$$

$$y_t = 3 + (-2)^t.$$

Por lo tanto, la solución no converge y oscila, ya que $|-2| > 1$ y $-2 < 0$.

$$(c) \begin{cases} 2y_t + \frac{1}{4}y_{t-1} = 5 \\ y_0 = 2 \end{cases}.$$

$$2y_t = \frac{-1}{4}y_{t-1} + 5$$

$$y_t = \frac{\frac{-1}{4}y_{t-1} + 5}{2}$$

$$y_t = \frac{-1}{8}y_{t-1} + \frac{5}{2}.$$

$$y_1 = \frac{-1}{8}y_0 + \frac{5}{2}$$

$$y_1 = \frac{-1}{8} * 2 + \frac{5}{2}$$

$$y_1 = \frac{-1}{4} + \frac{5}{2}$$

$$y_1 = \frac{9}{4}.$$

$$y_2 = \frac{-1}{8}y_1 + \frac{5}{2}$$

$$y_2 = \frac{-1}{8} \left(\frac{-1}{8}y_0 + \frac{5}{2} \right) + \frac{5}{2}$$

$$y_2 = \frac{1}{64}y_0 - \frac{5}{16} + \frac{5}{2}$$

$$y_2 = \frac{1}{64}y_0 + \frac{35}{16}$$

$$y_2 = \frac{1}{64} * 2 + \frac{35}{16}$$

$$y_2 = \frac{1}{32} + \frac{35}{16}$$

$$y_2 = \frac{71}{32}.$$

$$y_3 = \frac{-1}{8}y_2 + \frac{5}{2}$$

$$y_3 = \frac{-1}{8} \left(\frac{1}{64}y_0 + \frac{35}{16} \right) + \frac{5}{2}$$

$$y_3 = \frac{-1}{512}y_0 - \frac{35}{128} + \frac{5}{2}$$

$$y_3 = \frac{-1}{512}y_0 + \frac{285}{128}$$

$$y_3 = \frac{-1}{512} * 2 + \frac{285}{128}$$

$$y_3 = \frac{-1}{256} + \frac{285}{128}$$

$$y_3 = \frac{569}{256}.$$

$$2c + \frac{1}{4}c = 5$$

$$\frac{9}{4}c = 5$$

$$c = \frac{5}{\frac{9}{4}}$$

$$c = \frac{20}{9}.$$

$$y_t - c = (y_0 - c) \left(\frac{-1}{8} \right)^t$$

$$y_t - \frac{20}{9} = \left(2 - \frac{20}{9} \right) \left(\frac{-1}{8} \right)^t$$

$$y_t - \frac{20}{9} = \frac{-2}{9} \left(\frac{-1}{8}\right)^t$$

$$y_t = \frac{20}{9} - \frac{2}{9} \left(\frac{-1}{8}\right)^t.$$

$$\lim_{t \rightarrow \infty} y_t = \frac{20}{9}.$$

Por lo tanto, la solución converge a $\frac{20}{9}$ y oscila, ya que $\left|\frac{-1}{8}\right| < 1$ y $\frac{-1}{8} < 0$.

$$(d) \begin{cases} y_t - y_{t-1} = 3 \\ y_0 = 5 \end{cases}.$$

$$y_t = y_{t-1} + 3.$$

$$y_1 = y_0 + 3$$

$$y_1 = 5 + 3$$

$$y_1 = 8.$$

$$y_2 = y_1 + 3$$

$$y_2 = y_0 + 3 + 3$$

$$y_2 = y_0 + 6$$

$$y_2 = 5 + 6$$

$$y_2 = 11.$$

$$y_3 = y_2 + 3$$

$$y_3 = y_0 + 6 + 3$$

$$y_3 = y_0 + 9$$

$$y_3 = 5 + 9$$

$$y_3 = 14.$$

$$y_t = 5 + 3t.$$

$$\lim_{t \rightarrow \infty} y_t = +\infty.$$

Por lo tanto, la solución no converge y no oscila (diverge hacia $+\infty$), ya que $|3| > 1$ y $3 > 0$.

Ejercicio 4.

Encontrar todas las soluciones de las siguientes ecuaciones:

(a) $y_{t+2} - y_{t+1} + \frac{1}{2} y_t = 2.$

(b) $y_{t+2} - 4y_{t+1} + 4y_t = 7.$

(c) $2y_{t+2} + y_{t+1} - y_t = 10.$

(d) $y_{t+2} - 2y_{t+1} + 3y_t = 4.$

Ejercicio 5.

Encontrar las soluciones de las siguientes ecuaciones en diferencias y determinar si las soluciones convergen u oscilan en $t = \infty$:

$$(a) \begin{cases} y_{t+2} + 3y_{t+1} - \frac{7}{4}y_t = 9 \\ y_0 = 6 \\ y_1 = 3 \end{cases}.$$

$$(b) \begin{cases} y_{t+2} - 2y_{t+1} + 2y_t = 1 \\ y_0 = 3 \\ y_1 = 4 \end{cases}.$$

$$(c) \begin{cases} y_{t+2} - y_{t+1} + \frac{1}{4}y_t = 2 \\ y_0 = 4 \\ y_1 = 7 \end{cases}.$$

$$(d) \begin{cases} 2y_{t+2} + 2y_{t+1} + y_t = 2^{-t} \\ y_0 = 0 \\ y_1 = 0 \end{cases}.$$

Ejercicio 6.

Encontrar todas las soluciones de las siguientes ecuaciones:

(a) $y_{t+2} + 2y_{t+1} + y_t = 3^t$.

(b) $y_{t+2} - 5y_{t+1} - 6y_t = 2 \cdot 3^t$.

(c) $3y_{t+2} + 9y_t = 3 \cdot 4^t$.

(d) $y_{t+2} + 5y_{t+1} + 2y_t = e^t$.

(e) $y_{t+2} - 2y_{t+1} + 5y_t = t$.

(f) $y_{t+2} - 2y_{t+1} + 5y_t = 4 + 2t$.

(g) $y_{t+2} + 5y_{t+1} + 2y_t = 18 + 6t + 8t^2$.

(h) $y_{t+2} + 5y_{t+1} + 2y_t = e^t + 18 + 6t + 8t^2$.

Ejercicio 7 (*).

Resolver la siguiente ecuación en diferencia de orden 2:

$$y_{t+2} - 3y_{t+1} - 4y_t = 2^t + t^3.$$

Se parte de saber que la solución general de una ecuación en diferencias son las soluciones de la ecuación homogénea asociada a ésta más la solución particular:

$$y_t = y_t^h + y_t^p.$$

Siendo así, en primer lugar, se buscan las soluciones de la ecuación homogénea asociada a la ecuación en diferencias:

$$y_{t+2} - 3y_{t+1} - 4y_t = 0.$$

Sea $y_t = r^t$, se tiene:

$$\begin{aligned} r^{t+2} - 3r^{t+1} - 4r^t &= 0 \\ r^t (r^2 - 3r - 4) &= 0. \end{aligned}$$

Resolviendo mediante el método de Bhaskara, se hallan las soluciones de la ecuación homogénea:

$$\begin{aligned} r_1, r_2 &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \cdot 1 \cdot (-4)}}{2 \cdot 1} \\ r_1, r_2 &= \frac{3 \pm \sqrt{9+16}}{2} \\ r_1, r_2 &= \frac{3 \pm \sqrt{25}}{2} \\ r_1, r_2 &= \frac{3 \pm 5}{2} \\ r_1 &= \frac{3+5}{2} = \frac{8}{2} = 4. \\ r_2 &= \frac{3-5}{2} = \frac{-2}{2} = -1. \end{aligned}$$

Entonces, la solución homogénea es la siguiente:

$$y_t^h = C_1 4^t + C_2 (-1)^t.$$

En segundo lugar, dado que 1 no es raíz de la ecuación $r^2 - 3r - 4 = 0$, se propone la siguiente solución particular:

$$y_t^p = k_1 2^t + k_2 t^3 + k_3 t^2 + k_4 t + k_5.$$

Resolviendo en la ecuación en diferencias, se tiene:

$$\begin{aligned} &[k_1 2^{t+2} + k_2 (t+2)^3 + k_3 (t+2)^2 + k_4 (t+2) + k_5] - \\ &3 [k_1 2^{t+1} + k_2 (t+1)^3 + k_3 (t+1)^2 + k_4 (t+1) + k_5] - \\ &4 (k_1 2^t + k_2 t^3 + k_3 t^2 + k_4 t + k_5) = \end{aligned}$$

$$[k_1 2^{t+2} + k_2 (t^3 + 6t^2 + 12t + 8) + k_3 (t^2 + 4t + 4) + k_4 t + 2k_4 + k_5] -$$

$$3 [k_1 2^{t+1} + k_2 (t^3 + 3t^2 + 3t + 1) + k_3 (t^2 + 2t + 1) + k_4 t + k_4 + k_5] -$$

$$(4k_1 2^t + 4k_2 t^3 + 4k_3 t^2 + 4k_4 t + 4k_5) =$$

$$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5 -$$

$$3 [k_1 2^{t+1} + k_2 t^3 + 3k_2 t^2 + 3k_2 t + k_2 + k_3 t^2 + 2k_3 t + k_3 + k_4 t + k_4 + k_5] -$$

$$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 =$$

$$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5 -$$

$$(3k_1 2^{t+1} + 3k_2 t^3 + 9k_2 t^2 + 9k_2 t + 3k_2 + 3k_3 t^2 + 6k_3 t + 3k_3 + 3k_4 t + 3k_4 + 3k_5) -$$

$$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$$

$$k_1 2^{t+2} + k_2 t^3 + 6k_2 t^2 + 12k_2 t + 8k_2 + k_3 t^2 + 4k_3 t + 4k_3 + k_4 t + 2k_4 + k_5 -$$

$$3k_1 2^{t+1} - 3k_2 t^3 - 9k_2 t^2 - 9k_2 t - 3k_2 - 3k_3 t^2 - 6k_3 t - 3k_3 - 3k_4 t - 3k_4 - 3k_5 -$$

$$4k_1 2^t - 4k_2 t^3 - 4k_3 t^2 - 4k_4 t - 4k_5 = 2^t + t^3$$

$$-6k_1 2^t - 6k_2 t^3 - (3k_2 + 6k_3) t^2 + (3k_2 - 2k_3 - 6k_4) t + (5k_2 + k_3 - k_4 - 6k_5) = 2^t + t^3$$

Luego, se tiene que las constantes k_1 , k_2 , k_3 , k_4 y k_5 , respectivamente, son iguales a:

$$-6k_1 2^t = 2^t$$

$$k_1 = \frac{2^t}{-6 \cdot 2^t}$$

$$k_1 = \frac{-1}{6}.$$

$$-6k_2 t^3 = t^3$$

$$k_2 = \frac{t^3}{-6t^3}$$

$$k_2 = \frac{-1}{6}.$$

$$-(3k_2 + 6k_3) t^2 = 0$$

$$3k_2 + 6k_3 = \frac{0}{-t^2}$$

$$3k_2 + 6k_3 = 0$$

$$6k_3 = -3k_2$$

$$k_3 = \frac{-3k_2}{6}$$

$$k_3 = \frac{-1}{2} k_2$$

$$k_3 = \frac{-1}{2} \left(\frac{-1}{6} \right)$$

$$k_3 = \frac{1}{12}.$$

$$(3k_2 - 2k_3 - 6k_4) t = 0$$

$$3k_2 - 2k_3 - 6k_4 = \frac{0}{t}$$

$$3k_2 - 2k_3 - 6k_4 = 0$$

$$6k_4 = 3k_2 - 2k_3$$

$$k_4 = \frac{3k_2 - 2k_3}{6}$$

$$k_4 = \frac{1}{2} k_2 - \frac{1}{3} k_3$$

$$k_4 = \frac{1}{2} \left(\frac{-1}{6} \right) - \frac{1}{3} \frac{1}{12}$$

$$k_4 = \frac{-1}{12} - \frac{1}{36}$$

$$k_4 = \frac{-1}{9}.$$

$$5k_2 + k_3 - k_4 - 6k_5 = 0$$

$$6k_5 = 5k_2 + k_3 - k_4$$

$$k_5 = \frac{5k_2 + k_3 - k_4}{6}$$

$$k_5 = \frac{5}{6}k_2 + \frac{1}{6}k_3 - \frac{1}{6}k_4$$

$$k_5 = \frac{5}{6}\left(\frac{-1}{6}\right) + \frac{1}{6}\frac{1}{12} - \frac{1}{6}\left(\frac{-1}{9}\right)$$

$$k_5 = \frac{-5}{36} + \frac{1}{72} + \frac{1}{54}$$

$$k_5 = \frac{-1863}{17496}.$$

Entonces, la solución particular es la siguiente:

$$y_t^p = \frac{-1}{6} 2^t - \frac{1}{6} t^3 + \frac{1}{12} t^2 - \frac{1}{9} t - \frac{1863}{17496}.$$

Por lo tanto, la solución general de la ecuación en diferencias de orden 2 es:

$$y_t = C_1 4^t + C_2 (-1)^t - \frac{1}{6} 2^t - \frac{1}{6} t^3 + \frac{1}{12} t^2 - \frac{1}{9} t - \frac{1863}{17496}.$$